

Baibhab Karmakar

Master's Student in Earth Sciences, IISER Kolkata, Kolkata, India
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RESEARCH INTERESTS

Over the past two years, I have worked on earthquake source characterization, slip inversion, surface-wave amplification, and machine learning-based earthquake relocation, using observational and computational approaches to analyse seismic data. I aim to deepen my expertise through PhD research in understanding earthquake processes, seismic wave propagation, and Earth structure, with a particular interest in computational geophysics and data-driven methods in seismology.

EDUCATION

Indian Institute of Science Education and Research (IISER) Kolkata, Kolkata, India 2026 – Present

Master's Student in Earth Sciences

Thesis Title: Earthquake Relocation Using Machine Learning

Indian Institute of Science Education and Research (IISER) Kolkata, Kolkata, India 2022 – 2026
BS-MS, Earth Sciences Major and Computer Science Minor CGPA: 8.09/10.00

Higher Secondary Examination 2022
Score: 95.8%

Madhyamik Examination 2020
Score: 96.29%

RESEARCH EXPERIENCE

Computational Seismology Laboratory, IISER Kolkata Kolkata, India
Master's Researcher, under Prof. Supriyo Mitra 2026 — Present

- Applying QSeek, a machine-learning-based earthquake detection and localization framework, to continuous seismic data from the Kashmir Himalaya.
- Combining neural-network phase picking (via SeisBench pickers such as PhaseNet/EQTransformer) with stacking-migration and adaptive octree localization to build a data-driven earthquake catalog for the region.
- Benchmarking detection completeness and location accuracy of the QSeek-derived catalog against the existing regional catalog to characterize seismicity patterns in the Kashmir Himalaya.

Institut de Physique du Globe de Paris (IPGP) Paris, France
Summer Research Intern, under Dr. Martin Vallée June 2026 — July 2026

- Performed joint kinematic slip inversions for the 2015 and 2025 Charlie-Gibbs Transform Fault earthquakes using tele-seismic P/SH waveforms and moment-constrained Empirical Green's Function relative source time functions (RSTFs), resolving unilateral westward rupture propagation in both events.
- Used RBF interpolation and polygon overlap analysis to quantify asperity overlap and disjointness between the two ruptures at different slip thresholds.
- Revealed contrasting rupture patterns on the Charlie-Gibbs fault, with the 2015 earthquake dominated by a localized high-slip asperity, while the 2025 event exhibited more distributed slip and limited re-slip at the 2015 peak-slip region.
- Improved the seismic-data fit by incorporating spatially variable fault dip as an inversion parameter, while noting that the resulting dip distribution lacks independent observational constraints.

IISER Pune Pune, India
Research Intern 2025

- Investigated surface-wave amplification across the Shillong Plateau–Bengal Basin boundary using 1D shear-wave velocity profiles, focusing on the effects of low shear velocities and impedance contrasts at the basin edge.
- Modeled surface-wave propagation using SWRT (semi-analytical solutions across transversely aligned vertical discontinuities, incorporating higher-mode conversion) alongside SPECFEM2D full-waveform simulations; both methods predicted $2\text{--}5\times$ higher amplitudes in the basin than in crystalline terrain over 0.01–0.15 Hz.
- Validated model predictions against a regional earthquake recorded at both sites, finding good agreement on the transverse component; presented as a poster at AGU Annual Meeting 2025.

Environmental Science Laboratory, IISER Kolkata Kolkata, India
Research Assistant 2024

- Assisted in environmental data collection and analysis on pesticide contamination in milk samples.

TEACHING EXPERIENCE

IISER Kolkata

Teaching Assistant - Geophysics and Hydrology

Kolkata, India
Autumn 2026 - Present

- Assisting in tutorials and practical sessions for undergraduate students.
- Supporting students in conceptual understanding and problem-solving in geophysics and hydrology.

IISER Kolkata

Teaching Assistant - Introduction to Earth Science

Kolkata, India
Autumn 2025

- Assisted in conducting tutorials and practical sessions for undergraduate students.
- Supported grading, doubt-clearing sessions, and conceptual understanding of core Earth Science topics.

PROJECTS

Receiver Function Tool

Self-directed Computational Project

IISER Kolkata, India
2025

- Built an automated and interactive tool for crustal thickness estimation using receiver-function analysis.
- Developed workflows for seismic interpretation and theoretical travel-time modelling.

PUBLICATIONS

Journal paper

In Preparation

- Karmakar, B., Vallée, M.. Ten-year fault reactivation on the Charlie-Gibbs Transform Fault: joint kinematic inversion and slip asperity distribution analysis of the 2015 and 2025 earthquakes (working title, in preparation).

PRESENTATIONS

Poster Presentation - AGU Annual Meeting 2025

American Geophysical Union (AGU)

- Presented my work on seismic surface wave amplification in the Bengal Basin.
- Discussed observational and modelling results with an international research audience.

SELECTED COURSES

Master's Courses

- Master's Thesis Research in Computational Seismology

Additional Courses

- Data Science and Machine Learning
- Artificial Intelligence for Data Science

Bachelor's Courses

- Seismology
- Seismology Lab
- Geophysics
- Waves and Optics
- Structural Geology
- Geodynamics

AWARDS & FELLOWSHIPS

Innovation in Science Pursuit for Inspired Research (INSPIRE)

Department of Science and Technology, Government of India

India
2022 – Present

JBNSTS Junior Fellow

Jagadis Bose National Science Talent Search

India
2020 – 2022

SKILLS

- **Programming:** Python, C, MATLAB, Shell scripting
- **Python Libraries:** NumPy, Pandas, SciPy, Matplotlib, ObsPy, SeisBench, Sklearn, TensorFlow and etc.
- **Geophysical Software:** SAC, SPECFEM2D, SWRT, ASPECT(basic), Pyrocko
- **Machine Learning:** Supervised and unsupervised learning, data-driven modelling
- **Other:** Data Structures and Algorithms, Git, LaTeX, Microsoft Office Suite and etc.

OTHER EXPERIENCES

Library Management System

Software Project

IISER Kolkata, India

2024

- Developed an interactive library management system in C.
- Implemented book tracking and user management functionality.

Yuva Sangam Program (Phase 4)

Participant

Hariyan, India

2024

REFERENCES

Prof. Supriyo Mitra

Faculty, Department of Earth Sciences, IISER Kolkata, Kolkata, India

E-mail: supriyomitra@iiserkol.ac.in

Scholar Profiles: [Personal Page](#) - [Google Scholar](#)

Dr. Martin Vallée

Researcher, Institut de Physique du Globe de Paris (IPGP), Paris, France

E-mail: vallee@ipgp.fr

Scholar Profiles: [Personal Page](#) - [Google Scholar](#)

Dr. Arjun Datta

Assistant Professor, Department of Earth and Climate Science, Indian Institute of Science Education and Research (IISER) Pune, Pune, India

E-mail: arjundatta@iiserpune.ac.in

Scholar Profiles: [Personal Page](#) — [Google Scholar](#)